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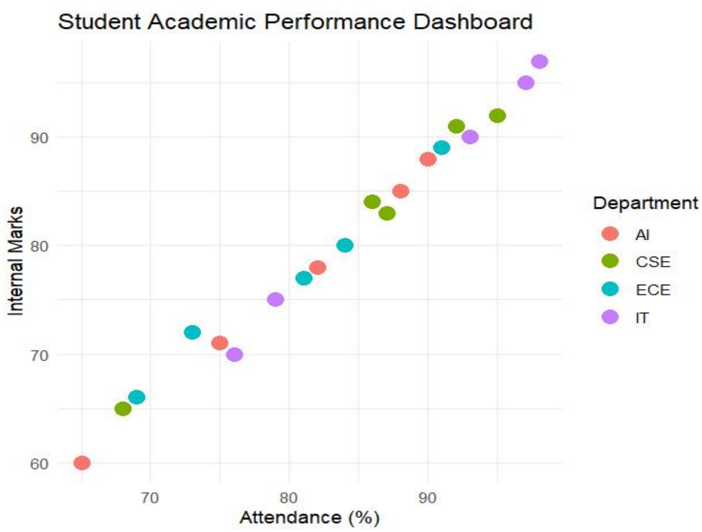
Scenario :1 Student Academic Performance Dashboard

Code:

```
> library(ggplot2)
> student_data <- read.csv("student_performance.csv")
  Student_ID Attendance Internal_Marks Department
> head(student_data)
 Student_ID Attendance Internal_Marks Department
1      S101         95             92        CSE
2      S102         88             85         AI
3      S103         76             70         IT
4      S104         91             89        ECE
5      S105         68             65        CSE
6      S106         82             78         AI
> ggplot(student_data,
+        aes(x = Attendance,
+            y = Internal_Marks,
+            color = Department)) +
+     geom_point(size = 4) +
+
+     labs(
+        title = "Student Academic Performance Dashboard",
+        x = "Attendance (%)",
+        y = "Internal Marks",
+        color = "Department"
+     ) +
+     theme_minimal()
```

Output:

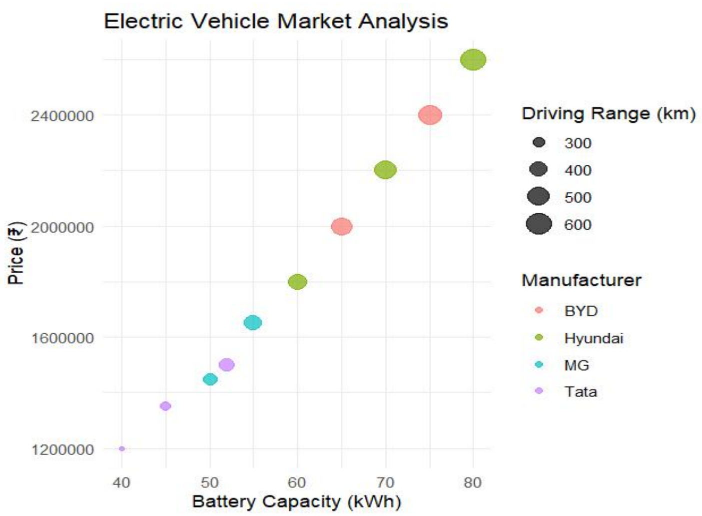


Scenario : 2 Electric Vehicle Market Analysis (R Studio)

Code:

```
library(ggplot2)
> ev_data <- read.csv("electric_vehicle.csv")
> head(ev_data)
  Vehicle_Model Battery_Capacity   Price Driving_Range Manufacturer
1       Model A               40 1200000           250         Tata
2       Model B               50 1450000           320           MG
3       Model C               60 1800000           420      Hyundai
4       Model D               45 1350000           280         Tata
5       Model E               55 1650000           380           MG
6       Model F               70 2200000           500      Hyundai
> ggplot(ev_data,
+        aes(x = Battery_Capacity,
+            y = Price,
+            size = Driving_Range,
+            color = Manufacturer)) +
+
+     geom_point(alpha = 0.7) +
+
+     labs(
+        title = "Electric Vehicle Market Analysis",
+        x = "Battery Capacity (kWh)",
+        y = "Price (₹)",
+        size = "Driving Range (km)",
+        color = "Manufacturer"
+     ) +
+
+     theme_minimal()
```

Output:

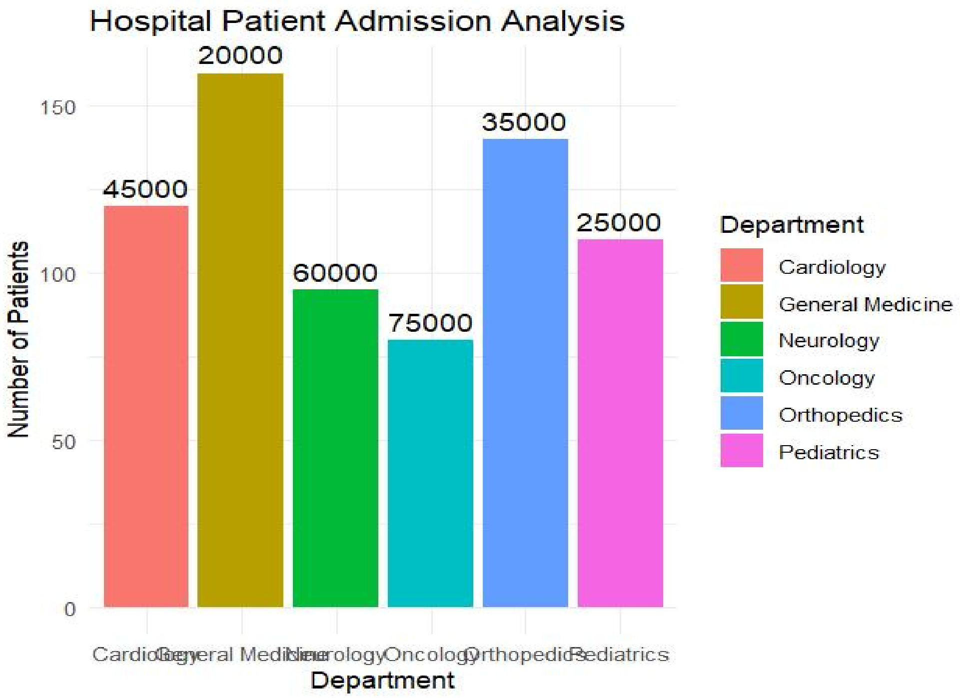


Scenario : 3 Hospital Patient Admission Analysis

Code:

```
library(ggplot2)
hospital_data <- read.csv("hospital_patient_admissions.csv")
print(hospital_data)
head(hospital_data)
ggplot(hospital_data,
      aes(x = Department,
          y = Number_of_Patients,
          fill = Department)) +
  geom_bar(stat = "identity") +
  geom_text(aes(label = Average_Treatment_Cost),
           vjust = -0.5,
           size = 4) +
  labs(
    title = "Hospital Patient Admission Analysis",
    x = "Department",
    y = "Number of Patients"
  ) +
  theme_minimal()
```

Output:

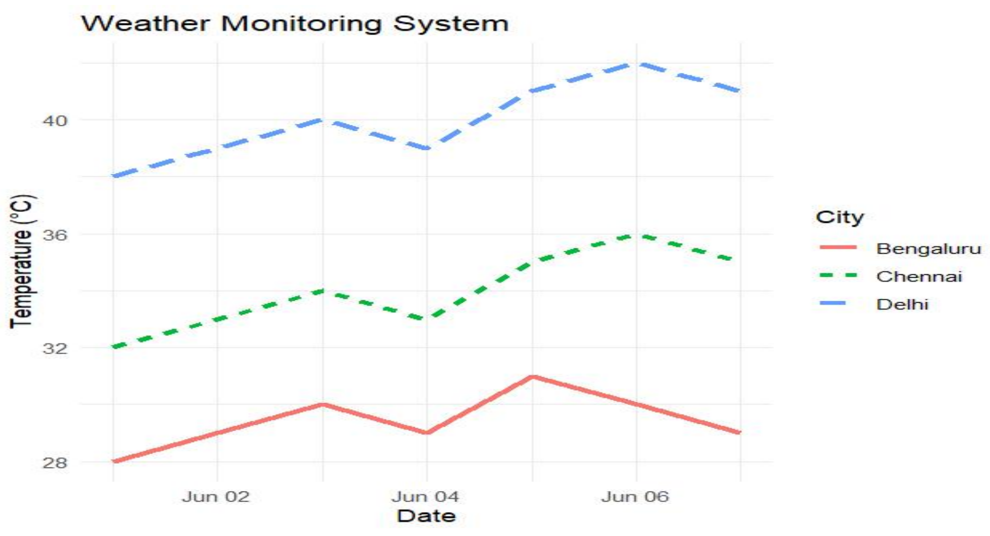


Scenario : 4 Weather Monitoring System

Code:

```
> library(ggplot2)
>
> weather_data <- read.csv("weather_monitoring.csv")
>
> weather_data$Date <- as.Date(weather_data$Date)
> head(weather_data)
        Date Temperature    City
1 2026-06-01          32 Chennai
2 2026-06-02          33 Chennai
3 2026-06-03          34 Chennai
4 2026-06-04          33 Chennai
5 2026-06-05          35 Chennai
6 2026-06-06          36 Chennai
>
> ggplot(weather_data,
+        aes(x = Date,
+            y = Temperature,
+            color = City,
+            linetype = City,
+            group = City)) +
+     geom_line(size = 1.2) +
+     labs(
+         title = "Weather Monitoring System",
+         x = "Date",
+         y = "Temperature (°C)"
+     ) +
+     theme_minimal()
```

Output:

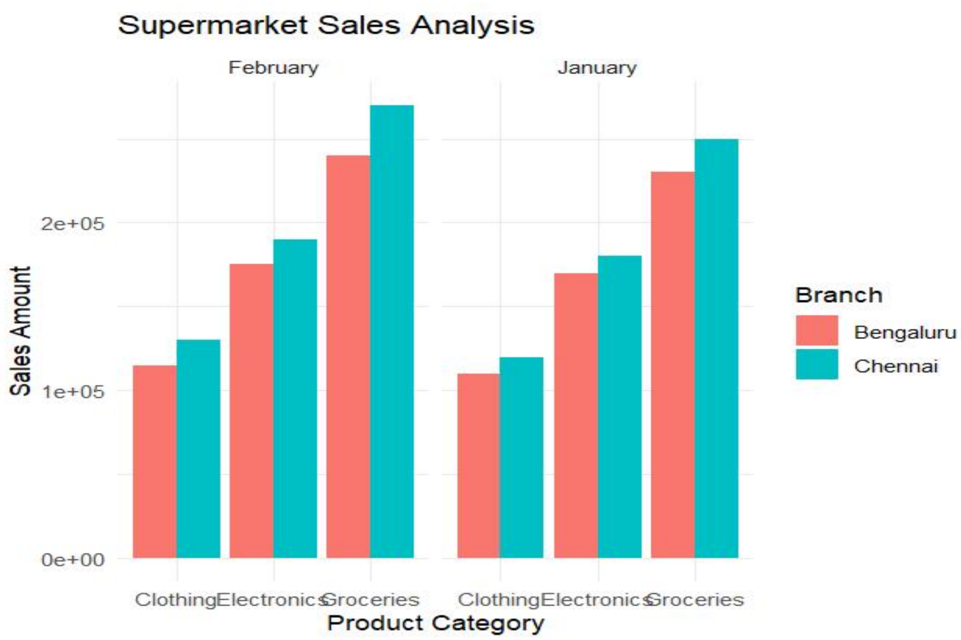


Scenario : 5 Supermarket Sales Analysis

Code:

```
> library(ggplot2)
>
> sales_data <- read.csv("supermarket_sales.csv")
> head(sales_data)
   Month Product_Category Sales_Amount    Branch
1 January        Groceries       250000   Chennai
2 January      Electronics       180000   Chennai
3 January         Clothing       120000   Chennai
4 January        Groceries       230000 Bengaluru
5 January      Electronics       170000 Bengaluru
6 January         Clothing       110000 Bengaluru
>
> ggplot(sales_data,
+        aes(x = Product_Category,
+            y = Sales_Amount,
+            fill = Branch)) +
+     geom_bar(stat = "identity", position = "dodge") +
+     facet_wrap(~Month) +
+     labs(
+         title = "Supermarket Sales Analysis",
+         x = "Product Category",
+         y = "Sales Amount"
+     ) +
+     theme_minimal()
```

Output:

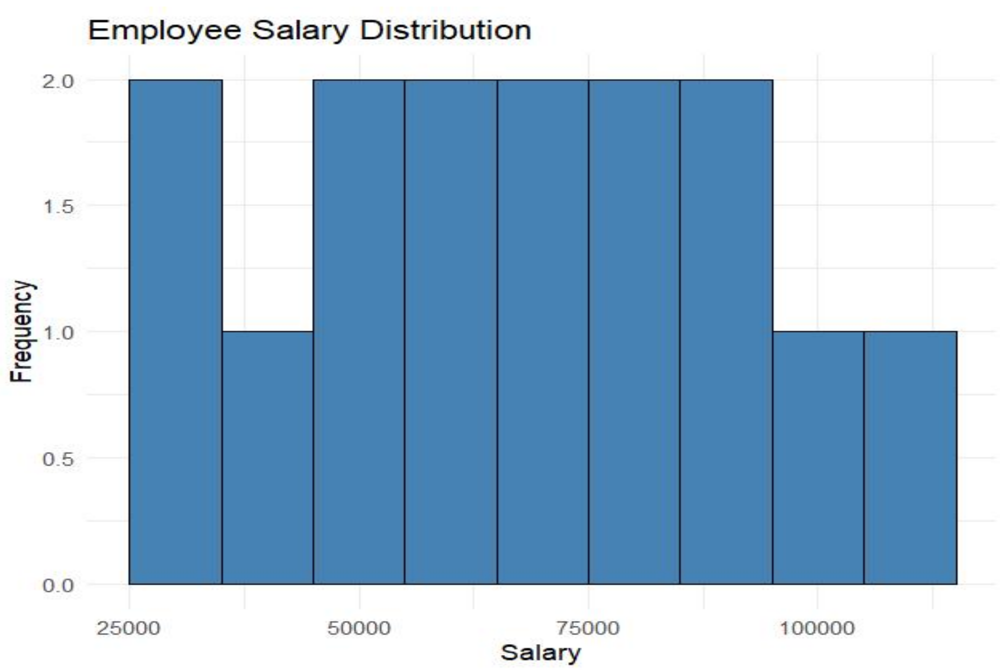


Scenario : 6 Employee Salary Distribution

Code:

```
library(ggplot2)
>
> salary_data <- read.csv("employee_salary_distribution.csv")
>
> head(salary_data)
  Employee_ID Salary Experience Department
1        E101  30000          1         HR
2        E102  35000          2    Finance
3        E103  42000          3         IT
4        E104  48000          4      Sales
5        E105  52000          5         HR
6        E106  58000          6    Finance
>
> ggplot(salary_data, aes(x = Salary)) +
+     geom_histogram(binwidth = 10000,
+                    fill = "steelblue",
+                    color = "black") +
+     labs(
+         title = "Employee Salary Distribution",
+         x = "Salary",
+         y = "Frequency"
+     ) +
+     theme_minimal()
>
```

Output:

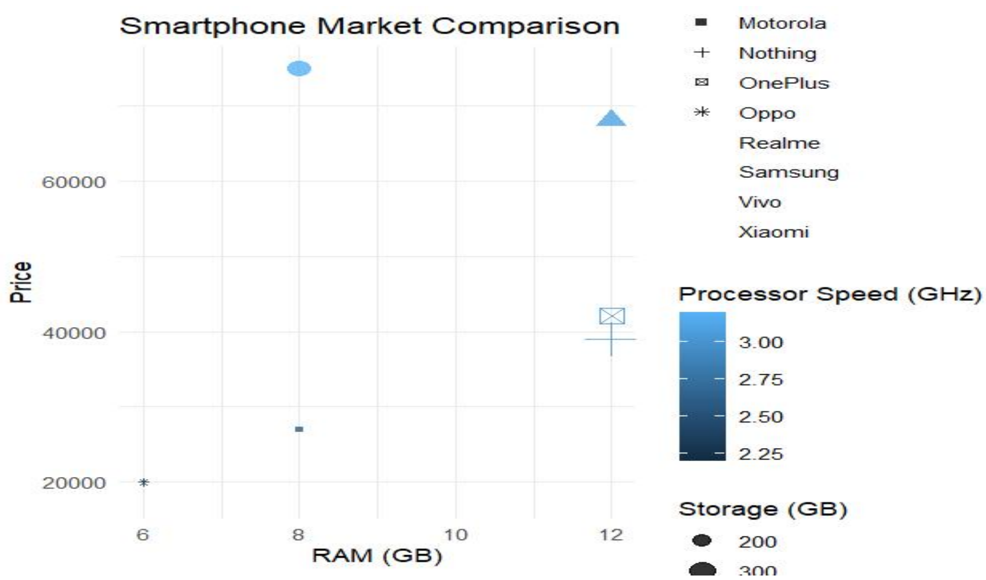


Scenario : 7 Smartphone Market Comparison

Code:

```
> library(ggplot2)
>
> phone_data <- read.csv("smartphone_market.csv")
>
> head(phone_data)
   Brand RAM Processor_Speed Price Storage
1 Samsung   8             2.4 25000     128
2  Xiaomi   6             2.2 18000     128
3 OnePlus  12             3.0 42000     256
4   Apple   8             3.2 75000     256
5  Realme   8             2.5 22000     128
6    Vivo  12             2.8 35000     256
>
> ggplot(phone_data,
+        aes(x = RAM,
+            y = Price,
+            shape = Brand,
+            size = Storage,
+            color = Processor_Speed)) +
+     geom_point(alpha = 0.8) +
+     labs(
+         title = "Smartphone Market Comparison",
+         x = "RAM (GB)",
+         y = "Price",
+         color = "Processor Speed (GHz)",
+         size = "Storage (GB)"
+     ) +
+     theme_minimal()
```

Output:

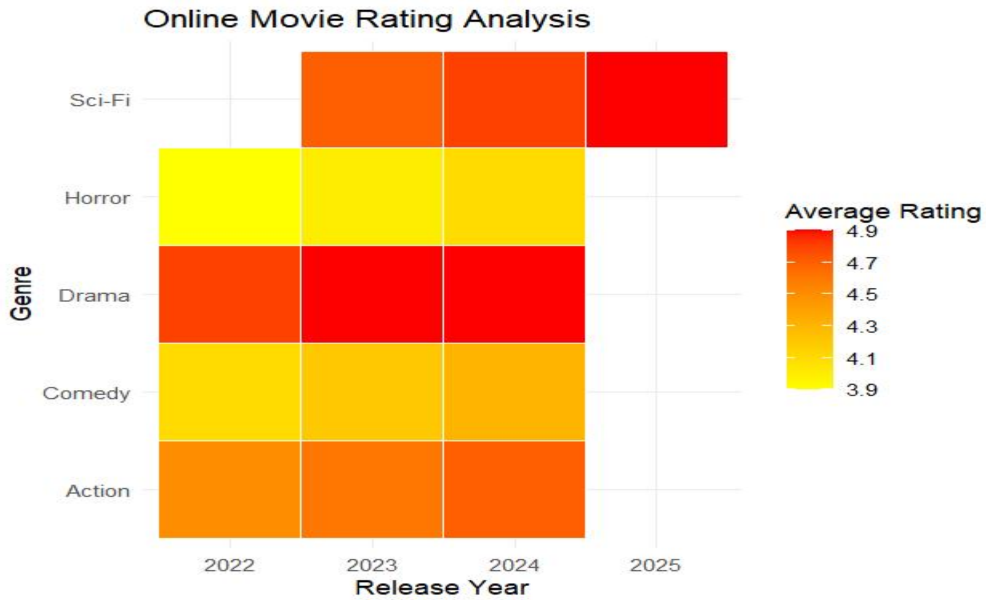


Scenario : 8 Online Movie Rating Analysis

Code:

```
> library(ggplot2)
>
> movie_data <- read.csv("movie_ratings.csv")
>
> head(movie_data)
  Genre Average_Rating Number_of_Reviews Release_Year
1 Action          4.5          12000        2022
2 Comedy           4.1           9500        2022
3  Drama          4.8          15000        2022
4 Horror          3.9           8000        2022
5 Sci-Fi          4.7          13500        2023
6 Action          4.6          14000        2023
>
> ggplot(movie_data,
+        aes(x = factor(Release_Year),
+            y = Genre,
+            fill = Average_Rating)) +
+     geom_tile(color = "white") +
+     scale_fill_gradient(low = "yellow", high = "red") +
+     labs(
+         title = "Online Movie Rating Analysis",
+         x = "Release Year",
+         y = "Genre",
+         fill = "Average Rating"
+     ) +
```

Output:

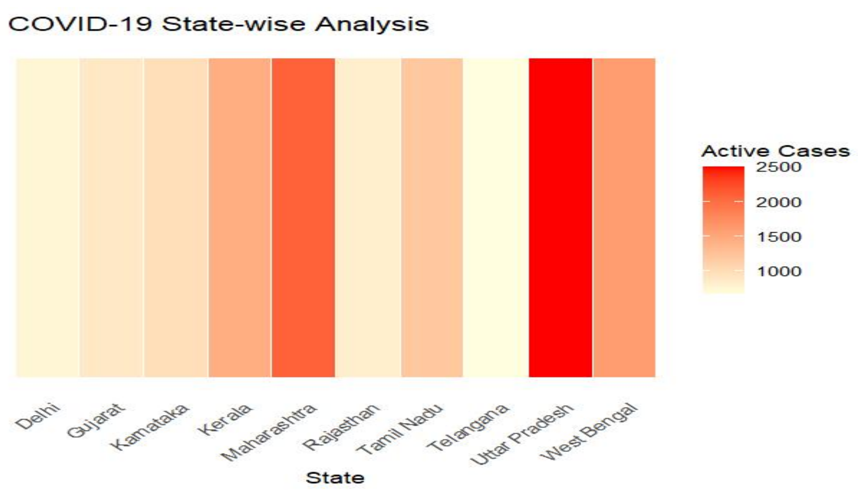


Scenario : 9 COVID-19 State-wise Analysis

Code:

```
> library(ggplot2)
>
> covid_data <- read.csv("covid_state_analysis.csv")
>
> head(covid_data)
        State Active_Cases Vaccination_Percentage Population
1  Tamil Nadu         1200                     94   78000000
2   Karnataka          980                     92   68000000
3      Kerala         1450                     96   36000000
4 Maharashtra         2100                     91  125000000
5       Delhi          760                     95   20000000
6     Gujarat          890                     90   71000000
>
> ggplot(covid_data,
+        aes(x = State,
+            y = 1,
+            fill = Active_Cases)) +
+    geom_tile(color = "white") +
+    scale_fill_gradient(low = "lightyellow", high = "red") +
+    labs(
+       title = "COVID-19 State-wise Analysis",
+       x = "State",
+       y = "",
+       fill = "Active Cases"
+    ) +
+    theme_minimal() +
+    theme(
+       axis.text.y = element_blank(),
+       axis.ticks.y = element_blank(),
+       panel.grid = element_blank(),
+       axis.text.x = element_text(angle = 45, hjust = 1)
+    )
```

Output:



Scenario : 10 Iris Flower Classification

Code:

```
library(ggplot2)

iris_data <- read.csv("iris_classification.csv")

head(iris_data)

ggplot(iris_data,
      aes(x = Sepal_Length,
          y = Petal_Length,
          color = Species,
          shape = Species,
          size = Sepal_Width)) +
  geom_point(alpha = 0.8) +
  labs(
    title = "Iris Flower Classification",
    x = "Sepal Length",
    y = "Petal Length",
    size = "Sepal Width"
  ) +
  theme_minimal()
```

Output: